

Behind the Curtains: Comparing Mozilla Hubs with Microsoft Teams in a Guided Virtual Theatre Experience

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ABSTRACT

This paper examines the perceived social presence experienced in social virtual environments (Mozilla Hubs) in a theatre context compared to video-based meeting platforms (MS Teams). To achieve this, we carried out a comparative study, designed a concept for a guided tour, recreated a theatre stage in a virtual reality environment, and granted students access to this stage as a replacement for visiting the location in person. Results show only minor differences between the conditions, which can be attributed to the difficult navigation control scheme provided by the devices (e.g., tablets, smartphones) and the lack of compelling onboarding features. Nonetheless, the proposed submission identifies design potentials of virtual social environments and offers insights into how the examined efforts are perceived both by the guides and students.

Index Terms: Virtual reality—Social virtual environments—Comparative study—Remote communication and collaboration

1 INTRODUCTION

In times of the *COVID-19* global pandemic, digital communication and collaboration platforms have gained significant importance. The current situation has left many companies to send their staff into remote working and home offices, rendering the possibility nearly impossible to have face-to-face meetings in person. This also applies to the culture and education sector, such as theatres, where various communication technologies were used to contact the target groups.

Video meetings embedded in *graphical user interfaces (GUIs)* are the most common workaround for addressing this problem. Such meetings frequently take place in various applications, such as *Zoom*¹ and *Microsoft Teams*², where a webcam feed represents every participant for everyone in the meeting to see. In these scenarios, social interactions are limited: participants can discuss topics or add resources via a chat channel. However, *Social Virtual Environments (SVEs)*, such as *Mozilla Hubs*³, offer a different approach by providing an interactive 3D setting, where people are represented by virtual characters that inhabit the artificial environment [5]. We deem that these technologies can foster communicative acts and can be used in various settings. In our case, we created an SVE in the theatre context, where students aged 12 to 18 are introduced to the processes behind the stage's curtain (e.g., lighting and building the stage design). Within the SVE, the young users get to know the various responsibilities and activities of the theatre staff and create their stage via predefined virtual objects. To identify the potentials and drawbacks of the approach, we conducted a comparative study that

examines the perceived social presence [11, 12] in SVEs concerning video meetings embedded in a guided tour through a theatre.

The paper provides the following contributions: first, it identifies the potentials of SVEs in comparison to video meetings in the given context through a study. Second, the work provides insights into how the guides and the students perceive the examined efforts.

2 RELATED WORK

SVEs have already been subject to various studies, especially in regards to *Virtual Reality (VR)* [10, 19, 20, 23]. VR environments can enhance motivation and learning abilities under the beneficial usage of Social VR and gamification [16]. Many studies have shown that users can benefit from video game-like scenarios (e.g., [4, 6, 8, 9, 17, 18, 26]). Acquired skills apply to the real world and can also be used in the real world. Mozilla Hubs combines essential factors to render a balanced system that could be equally engaging as a video game. Represented as an avatar, users can visit conferences, festivals, galleries, attend classes, and much more [3, 13, 15, 21, 25].

To have a robust platform at hand is a desirable starting point if one is to create a SVE. But without the knowledge of how to engage users in a meaningful and exciting way, the attention span will not pull its necessary weight. Studies have gone out of their way to track, collect, and analyze data that can be harnessed to make competent decisions when creating Mozilla Hubs rooms [24] to pinpoint and observe the behavior of users.

The necessity of relying on technology for educational purposes has already born studies that aim at finding answers to similar questions like the one this paper attempts to answer. Tatli et al. [22] conducted a study called: *Learning via Virtual and Real Museums: A Comparative Study on Presence and Retention* in which 28 students were separated into two groups. The first group visited a museum, and the second half visited the same museum, but virtually. Interviews and drawings from the students were used to evaluate the students' learning effect. The results unveiled that the virtual group achieved a similar perceived presence than the students who physically visited the museum. Learning and knowledge retention statistics show a similar picture [22]. Various studies support these results. Similarly, virtual tours or even VR tours through museums or exhibitions can enhance the experience [1, 7, 14].

3 COMPARATIVE STUDY: A REMOTE GUIDED TOUR IN A THEATRE CONTEXT

In collaboration with the Landestheater Linz⁴, we aimed at identifying the potentials of SVEs in a theatre context in comparison to video-based meeting platforms (e.g., MS Teams). To achieve this, we carried out a comparative study and designed a concept for a guided tour, recreated a theatre stage in VR environment⁵, and granted schools access to this stage as a replacement for visiting the location in person (see figure 1).

Specifically, the study aimed to determine if pupils in a remotely guided theatre tour experience a higher sense of social presence than only seeing what is happening through a video chat. To measure

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¹<https://zoom.us/>

²<https://www.microsoft.com/en-wv/microsoft-teams/>

³<https://hubs.mozilla.com/>

⁴<https://www.landestheater-linz.at/>

⁵<https://hubs.mozilla.com/mVxJ98r/fhooe-hagenberg>

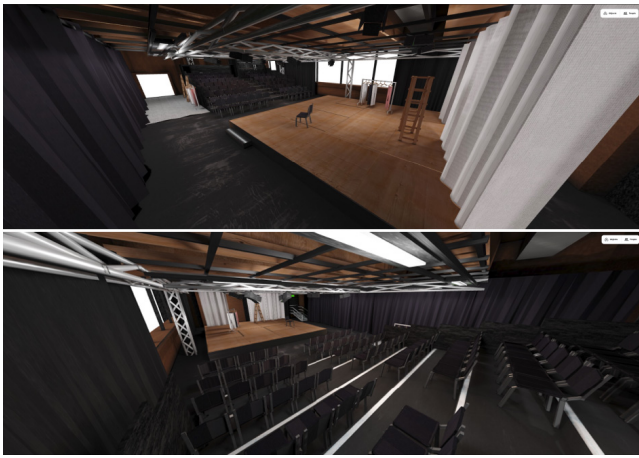


Figure 1: Screenshot of Mozilla Hubs theatre stage in VR.

this aspect, we employed a questionnaire that consists of various components: Apart from demographics (i.e., age, gender, experience with technology), it was essential to distinguish between participants who were able to participate actively and who could not (6 items). This distinction provides a balanced and direct comparison of those actively participating and those experiencing the same event but not actively. The actively participating students accessed the virtual stage through a screen-based medium (i.e., computer, notebook, or tablet). Students who could not participate actively because their hardware was inappropriate for Mozilla Hubs could join a MS Teams meeting. A guide from the theater entered the virtual stage and shared the screen on MS Teams. The guide took commands from the spectators. Together, they explored the virtual theater and interacted with the participants in Mozilla Hubs. This would substitute the experience as close as possible. Besides, students, who could not actively participate in Mozilla Hubs, filled out a different form. Here, it was crucial to understanding why some of the pupils could not participate if they encountered any technical difficulties and, if so, which solutions could be offered? Besides, we asked subjects if there were other reasons one might choose not to participate actively.

Social presence was measured via the *Networked Minds Social Presence Inventory* by Biocca and Harms [2] (9 items). Subjects could rate statements, such as “I often felt as if my school colleague and I were together in the same room” on a 5-point Likert scale. The questionnaire concluded with an open question where students could provide feedback to the questionnaire or the experience itself. This allowed for additional insight on topics that might have been missed by the form and more direct feedback on what to improve in further installments of similar rooms and experiences.

Concerning the procedure, the students were split into three groups of various sizes each. The groups were asked to create a stage design read to them by the group’s guide. To determine how students perceive the different forms of interaction (i.e., MS Teams vs. Mozilla Hubs), they could use MS Teams to instruct the guide to modify the scene or directly interact with the objects via Mozilla Hubs. Unbeknownst to them, the groups were all recreating parts of the scenes from Johann Wolfgang von Goethe’s tragic play *Faust*, and the guides were monitoring their behavior (see figure 2). After a particular time, all groups came together, and one speaker of each group was presenting the fruits of their labor. Additionally, every student was asked to fill out a questionnaire to gain deeper insight into the group’s behavior and thought process.



Figure 2: Stage design created by eight students at the age of 13 to 14 years.

4 RESULTS

Over three weeks, 36 students (13 male, 23 female, age M:14.1 and SD: 0,76) filled out the form and gave feedback on their experience in Mozilla Hubs and MS teams. Statistical analysis was carried out with Jamovi⁶. Mean values of each dimension indicate that there is no clear preference on whether the student was actively participating or watching the activity on a shared screen in a Microsoft Teams meeting (see table 1). This was also confirmed through a t-test: There was no significant difference on all three sub-scales (see table 2).

An important factor that has the potential to shift the view and interpretation of data is age. In both groups – active and passive – the vast majority of participants were 13, 15, or 18. Given these numbers, it is safe to assume that age in this particular case does not play as big of a role as one might expect. Another factor that can safely be ruled out is gender. There is no significant difference in perception of social presence, perceived presence, or social attention between female and male subjects.⁷ This result leaves room for discussion on why there is no apparent distinction between active and passive participation. Following this notion, several questions arise: Is there a reason why no significant findings have not been found? If not, is SVE a subject worth discussing? Were errors made in the process of this study that influenced the outcome of this questionnaire?

5 DISCUSSION

This study’s outcome led to some interesting results: It was anticipated that students who were at their homes over the last couple of months, due to the ongoing pandemic, would heavily appreciate the change of pace and the transposition from teacher-centered teaching to a hands-on experience in a SVE.

⁶<https://www.jamovi.org/>

⁷Participants were given the option to identify their gender as “diverse”. None of the participants choose this option

Table 1: Table showing the mean and standard deviation social presence values (co-presence, attentional engagement, perceived comprehension) per condition (Mozilla Hubs, MS teams).

Descriptives		Condition 1: Mozilla Hubs 2: Teams	Co-Presence	Attentional Engagement	Perceived Comprehension
N	1	22	22	22	
	2	14	14	14	
Mean	1	2.95	2.82	3.09	
	2	2.98	2.69	3.02	
Median	1	3.00	2.83	3.00	
	2	3.00	2.67	2.67	
Standard deviation	1	0.586	0.640	0.714	
	2	0.690	0.768	1.020	

Table 2: Table showing the t-tests concerning social presence (co-presence, attentional engagement, perceived comprehension).

Independent Samples T-Test					
			Statistic	df	p
Co-Presence	Student's t	-0.129	34.0	0.129	
Attentional Engagement	Student's t	0.540	34.0	0.593	
Perceived Comprehension	Student's t	0.233	34.0	0.817	

A substantial portion of students found it challenging to navigate through Mozilla Hubs, especially mobile devices. The comparative study's tutorial only focused on PCs and Notebooks, not tablets and smartphones. Frustration over the actual navigation most likely directly interfered with the students' experience. Suppose the participant is preoccupied with figuring out how to complete the most basic given tasks. In that case, there will be less capacity to pay attention to other classmates or the social presence this experience aims to deliver.

This observation was quite surprising since almost every participant has experience with video games. Due to the novelty of this technology in its current form, it is crucial to prepare users beforehand. One way of doing so is a tutorial that users can watch and learn from. Another vital measure to ensure a barrier-free entry into a SVE is an onboarding phase: This can happen in digital signs that are visible to users at spawn without the need to move or look around. These signs display the basic controls. As with most more prominent groups, there is no guarantee that every user will navigate through the experience. Not everyone will have watched the tutorial or noticed the signs in the onboarding section. The presence of a tech support team is highly advisable for bigger events. Such tech support can aid and assist virtual stranded avatars in experiencing a problem with navigation or technical sorts.

A short interview with the guides shed some more light on the topic. Communication would seem to be a key component in a group task where the goal is to achieve a unified goal. Yet, the guides report a total lack thereof. The students rarely used the built-in chat function, not verbally or text. If communication was happening, it was mainly looking for confirmation if the task was understood correctly and a general search of affirmation of the taken action. This behavior stands true to the active participants and the passive participants watching the shared screen of the guide on Microsoft Teams. The guide offered to take commands if expressed in voice or text chat, but such an event never occurred. The guides voiced their assumption that the students were not communicating via the

"monitored" channels like the Mozilla Hubs voice chat. There were signs of communication outside of these official channels. This claim, however, can not be backed and verified but raises the need of altering the questionnaire with additional questions such as "Which ways of communication did you use to communicate with your school colleagues?"

These two factors are not the only ones that explain why there is a lack of diversity in the questionnaire results. The majority of information given in the feedback section at the end of the questionnaire highlights another fact: the appreciation of a more than welcome change of format in teaching. Most of the students found this virtual field trip engaging, and the majority even stated that this experience had nourished their interest in theater.

6 CONCLUSION

This paper examined the perceived social presence experienced in virtual social environments (Mozilla Hubs, accessed with a screen-based medium) in a theatre context compared to video-based meeting platforms (MS Teams). Findings revealed only minor differences between the media, attributed to the problematic navigation control scheme provided by specific devices (e.g., tablets, smartphones) and the lack of compelling onboarding features. We carried out a comparative study and designed a guided tour concept to grant students access to this stage as a replacement for visiting the location in person. In a further step, the study could be extended to investigate how a screen-based experience compares to a fully immersive experience. We intend to explore other platforms to determine if the gathered results can be reproduced via other platforms concerning the future steps. Furthermore, we deem that further examining different target and age groups could be a worthwhile endeavor. Last but not least, other areas of application, such as museums or students spaces, could be investigated.

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